Niayesh Afshordi

Astrophysics, cosmology, and gravity

Professor, University of Waterloo; Research Associate Faculty, Perimeter Institute

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Short public CV — updated 30 October 2026

Profile

Niayesh Afshordi is an astrophysicist whose research connects fundamental questions in gravity and quantum physics to observations of the universe. His work spans black holes, gravitational waves, the early universe, the cosmic microwave background, dark matter and dark energy, cosmic structure, and observational tests of gravity, with interdisciplinary applications in medical physics and epidemiological modelling.

Appointments

Full Professor, Department of Physics and Astronomy, University of Waterloo — 2022–present

Research Associate Faculty, Perimeter Institute for Theoretical Physics — 2009–present

Founding Member, Waterloo Centre for Astrophysics — 2018–present

Education and previous appointments

PhD, Astrophysics, Princeton University — 2000–2004

BSc, Physics, Sharif University of Technology — 1996–1999

Distinguished Research Fellow, Perimeter Institute — 2007–2009

Postdoctoral Fellow, Institute for Theory and Computation, Harvard–Smithsonian CfA — 2004–2007

Selected research support

NSERC Discovery Grant: "Quantum Black Holes, Vacuum, and the Big Bang" — C$250,000 | 2025–2030

University of Waterloo Interdisciplinary Trailblazer Grant: "Quantum Spacetime in the Cosmos" — C$80,000 | 2022–2025

NSERC Discovery Grant: "Novel Astrophysical Windows into Fundamental Physics" — C$170,000 | 2020–2025

Consulting

Aligned — consulting on testing frontier AI models in scientific research — March 2026–present

Selected recognitions

Buchalter Cosmology Prize — 2019; 2015

Professor M. K. Vainu Bappu Gold Medal — 2011

Ontario Early Researcher Award — 2011

NSERC Discovery Accelerator Supplement — 2010–2013

Research group and scholarly record

Current group: Alice Chen, Ruolin Liu, Hassan Subhi, Shokoufe Faraji, Ohoud K. Alharbi, Harrison (Haoye) Li, Robin Oberfrank, Aliv Sahoo, Batia Friedman-Shaw, Noni Hajatri, Annika Cheng, and Owen Roth. Projects and alumni: `People`.

2026 preprints: quantum-gravity interferometry; DESI/gravity; primordial spectrum; LQC bounce; quadratic-gravity inflation; thermal SZ/unWISE; Big Mysteries Survey. All 156 arXiv papers and the full supervision/talk record: `Full CV`.

Recent invited talks: Kavli IPMU, 1 October 2026; University of New Mexico, 23 October 2026. Full titles and earlier talks appear in the Full CV.